Project Initiation Document

Personal Digital Pantry Tracker & Grocery List App

Requirements Document

Version 1.0

10/10/2025

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Project Initiation

INTRODUCTION

This document outlines the requirements, application purpose, and intended goal for the PantryPal app. A mobile app designed to assist users with the headache of managing their cupboards! The app will allow users to remotely keep track of what groceries are available within a pantry, and allow for easily constructed plan meals and reduce food waste. This document will outline the functional, non-functional, and security requirements involved in the development of the mobile app. Furthermore the document will also include the system objectives, user interaction, data handling, deliverables schedule, and other necessary information for the creation of the app.

This document is intended for the development team, project stakeholders, designers, quality assurance personnel, and CEO. It is intended to make clear all expectations and specifications desired by the customer to ensure that the app meets the outlined specifications.

Future revisions to this document may include: integration with online grocery API's, cloud based backup options, user account management, and AI categorization. Any updates will be noted and reflect any new functional features, bug fixes or security improvements as the app evolves.

PROBLEM DESCRIPTION

Cooking and keeping track of home pantry inventory requires a lot of mental load in having to organize the pantry but also having to scrounge around for the right ingredients. Even during grocery shopping, an essential part of re-stocking a pantry; customers have to find a way to write down every single food and produce to buy at once. This process is inefficient, and often feels like a chore; leading to potentially not purchasing, overpurchasing, and/or unnecessary food waste. The PantryPal app seeks to provide a centralized system to log/monitor home kitchen pantries. The app works both as an inventory tracker and a grocery tracker. By digitalizing a pantry, individuals have a clear menu to track what items are present without having to scrounge the physical pantry. When it's time to cook, the PantryPal app seamlessly subtracts portions of the inventory and is then added to a grocery list. The grocery list portion of the app can then be used for grocery shopping related tasks. The introduction of PantryPal will reduce the cognitive load of having to track food and pantry inventory for expiration dates, availability, and quantity.

DOCUMENT MANAGEMENT

Document versions will be managed in cloud storage, with versions saved after each major revision. Starting with PantryPal version 1.0, each bug fix would increment 0.01, each feature addition or change would increment 0.1, and each large version update would increment 1.0. Each previous version will be saved in case of need for rollback due to later bugs or the platform crashes. Only essential team members will view Document versions.

INTENDED AUDIENCE

The intended audience for the viewing of this document includes stakeholders: including Sponsors, developers, project managers, product managers, as well as end-users and professors overseeing the initiation document.

PROJECT GOAL(S)

The goal of the project is to create an efficient, automated service that bridges the gap between grocery shopping, meal planning, and pantry inventory management. To achieve this goal, the app needs to seamlessly populate the inventory and reduce manual inputs as much as possible. The app thus needs a connection to the internet to perform requests to an API OCR server to reduce manual inputting. No data is being extracted and sold for PantryPal.

PantryPal shines by being an easy to use app free to all users, but the tradeoff means that the product needs to be profitable to shareholders. Advertisements are essential in keeping the apps free while simultaneously being a source of revenue to justify investment.

PROJECT TECHNICAL GOALS

This project will be implemented as a stand alone mobile app. OCR integration is required to scan and extract data to show in the app. Responsive and understandable pantry management.

Security testing will need to be done during the development process to make sure that the only information being sent between the app and the API server only extracts information from the images being sent.

Functionality testing: There should be testing for proper functionality of the product, including the inventory portion, the grocery portion, and the proper readings of the API portion.

Edge case testing will verify that the application works as intended in every possible scenario. This will provide reliability and security to the application, preventing any unintended behavior.

BUSINESS PURPOSE

PantryPal will provide consumers with a seamless experience not offered by other applications. This app provides the utmost convenience for meal planning and grocery shopping, inviting users to worry less about deciding what they should make for dinner, having to think about the current state of the user's inventory, and having to mentally remember or manually write down an impromptu grocery list. Users will fall in love with the amount of automation and integration of a digital inventory, taking the menial task and turning it into a stress free experience.

PantryPal is also tapping into an undiscovered market with the goal being in simplicity, with the financial incentive being in having advertisements for the usage of a free app.

FUNCTIONAL PURPOSE

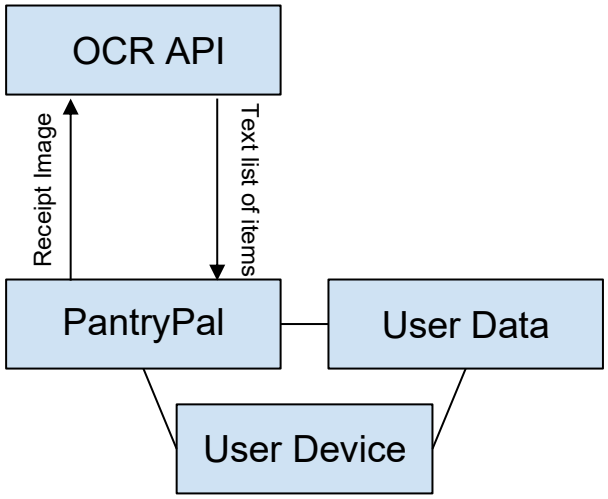
PantryPal will provide the ultimate pantry service, allowing for seamless integration between grocery shopping and meal planning.

STAKEHOLDERS

Sponsors, developers, project managers, product managers are involved in gauging the value of the product and the place the product stands in the market, as well as actively contributing to the inception, creation, development, deployment, and the maintenance of the app. End-users and professors have a primary stake in the implementation of the app by being the main targeted audience for the usage of the product, as well as gauging the feasibility of the product being deployed.

Secondary stakeholders (those who are indirectly impacted): App stores are involved for app distribution as well as the usage of 3rd party APIs for OCR functionality.

PROJECT DIAGRAMS



OUT OF SCOPE

1. **Cloud-based database** for user pantry data (all information will remain stored locally).
2. **Photo storage** beyond receipt scanning functionality.
3. **Spelling autocorrect** for manual item entry.
4. **Web compatibility** — the application will be mobile-only for this phase.
5. **Personal information storage** such as names, addresses, or account profiles.
6. **Health monitoring features** (e.g., blood pressure, heart rate, or weight tracking).
7. **Diet tracking or calorie counting** functionality.
8. **Voice recognition** or smart assistant integration.
9. **User accounts or login systems** — data will not be tied to personal identities.
10. **Cross-device synchronization** (no automatic syncing between devices).
11. **Recipe creation or upload features** (only preloaded suggestions will be supported).
12. **In-app purchases or premium tiers** (the app will remain free for the initial release).

13. **Push notifications or reminder systems.**

14. **Social media integration** (e.g., sharing pantry lists or meals online).

ASSUMPTIONS

- **Technical Assumptions:**
    - Mobile devices are required.
    - Mobile devices are compatible with the application.
    - Data related to the app is stored locally on the mobile device.
- **Test Assumptions:**
    - The OCR API will be able to recognize the receipts reliably and accurately.

CONSTRAINTS

- If developers are sick there will be delays.
- Certain laws or policies may impact development.
- OCR might not recognize receipts.
- Unable to transfer data from the photo to the app (unable to integrate).

RISKS

The following risks can create issues for the project. These risks may create issues that have an uncertain effect on the project which in turn effect achieving the desired results (e.g., satisfying requirements, meeting project goals and priorities, achieving measures of success, etc.) stated in this document.

PantryPal cares about user privacy and data security. All grocery and pantry data are stored locally by default, to minimize external data exposure. If cloud synchronization or online features are added in future versions, secure encryption protocols (e.g., AES-256) and user consent-based data sharing will be implemented to comply with data protection standards. Advertisements will not be targeted so no data is gathered from the user.

GLOBAL REQUIREMENTS

Global Standards

The system shall comply with all policies and standards regarding data protection and handling. Compatible with Android/iOS and able to be supported long-term. Explain why a camera is necessary (OCR) and possibly location. Allow for maintainability and improvement.

General

The system shall allow users to freely edit the content of inventory (adding, removing, editing). Notification of low stock and expiring items.

Design

The system shall be able to present all necessary information in a way that does not overwhelm or confuse the user. Consistent style and clear navigational features. Support different screen sizes/devices. Personalized items and lists able to be created.

Performance Requirements/Performance Engineering

- This system should be able to present the current inventory within seconds. 1000 hourly.
- After uploading the receipt picture the system shall process and input it within 15 seconds. 1000 hourly.
- Editing the pantry should run smoothly up to 10,000 items. 10,000 hourly.
- List generation will be updated any time there is a change within the pantry. 10,000 hourly.
- List generation will update daily to account for expiring items. Daily.

Security

The system shall securely handle all operations conducted within the app, leaving all data protected locally. There will be no user accounts.

Privacy

The system shall not share any personal information of the user in any way. All personal data will be stored locally, and any transmitted data will not relate to the user's personal information. Metadata from images used will be scrubbed to protect all user information.

Reporting Requirements

The system shall report popular recipes, performance values for various features, and users' utilization frequency. It will also return metrics on advertisements and user interaction with promoted meals/pantry items.

Other Non-Functional Requirements

The system shall feel automated, requiring little input from the user to update their pantry or receive meal recommendations. Many features will be presented, rather than requiring user input.

SUCCESS DELIVERABLES

- **October 10, 2025 – Project Initiation Document**
  Completion of the project initiation document, outlining scope, goals, and requirements.

- **October 13, 2025 – Project Start Date**
  Official start of development and team coordination activities.

- **December 20, 2025 – Base System Release**
  Initial product launch timed for the New Year's health wave. This version will include the core pantry management system, enabling users to easily track and manage their household inventory.

- **February 14, 2026 – Streamlined System Update**
  Feature update introducing meal suggestions based on available ingredients. Released around Valentine's Day, this update supports users planning meals for special occasions.

- **April 1, 2026 – Version 2.0 Release**
  Major update introducing potential cloud-based storage, meal sharing, and expanded user interaction features. Marks the baseline project completion, with an option to continue development based on performance and feedback.

Appendix A: Record of Changes

Table 4 - Record of Changes

Version Number	Date	Author/Owner	Description of Change
<1.0>	<10/10/2025>	Member 1	Introduction, and Success deliverables
<1.0>	<10/10/2025>	Alvin	Technical goals, project goals
<1.0>	<10/10/2025>	Member 2	Constraints
<1.0>	<10/10/2025>	Member 3	Functional purpose

Appendix E: Approvals

The undersigned acknowledge that they have reviewed the Requirements Document and agree with the information presented within this document. Changes to this Requirements Document will be coordinated with, and approved by, the undersigned, or their designated representatives.

Table 8 - Approvals

Document Approved By	Date Approved
Zoe Davenport, Head of Partnerships – MarketMesh Corp.	10/10/2025
Alex Mercer, Business Owner – ShelfSense Inc.	10/10/2025
Riley Park, Project Manager – PantryPal Ventures	10/10/2025
Evan Torres, Technical Lead - StockKeep Technologies	10/10/2025
Samira Khan, Stakeholder – ReceiptAI Co.	10/10/2025